

A Two-Plane Counterexample to the Atomic Condition for a Plücker-Coordinate ℓ^p Integrand

Automated mathematics run `fa_banach_001`, lane 6

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Status. Candidate counterexample, likely valid; prepared for expert review. The result disproves the all-exponent atomic-condition claim suggested in Remark 6.4 of De Rosa–Tione [1].

1 Source question

The source is Antonio De Rosa and Riccardo Tione, *Regularity for graphs with bounded anisotropic mean curvature*, arXiv:2011.09922v2, published in *Inventiones Mathematicae* 230 (2022), 463–507. On page 26, Remark 6.4 reports numerical evidence that the integrand on $\mathbb{G}(4, 2)$ induced by the ℓ^p norm of the six Plücker coordinates satisfies (SAC), hence (AC), for every $p \in (1, \infty)$, and states that no analytic proof was found.

This is impossible since $\|\tau\|^\epsilon = \|v_1 \wedge v_2\|^\epsilon = \sum_{i < j} |v_{ij}|^\epsilon = 1$ μ' -a.e., that is a consequence of the fact that μ is supported on G . Hence at least one of the S_{ij} must be invertible, as desired. $\square$

Remark 6.4. *Let us remark that the considerations at the beginning of the section and Lemmas 6.2-6.3 hold in $\mathbb{G}(N, m)$, but we have chosen here to use only $\mathbb{G}(4, 2)$ for ease of notation. We do not know whether Theorem 6.1 holds for higher dimensions or codimensions. Moreover, numerical simulations indicated that for $\mathcal{G}(\cdot) = \|\cdot\|_{\ell^p}$, $p \in (1, \infty)$, $\Psi_{\mathcal{G}}$ fulfills (SAC), hence actually satisfies (AC), by Proposition 3.5. Unfortunately, we were not able to prove it analytically.*

The precise sentence is: “numerical simulations indicated that for $\mathcal{G}(\cdot) = \|\cdot\|_{\ell^p}$, $p \in (1, \infty)$, $\Psi_{\mathcal{G}}$ fulfills (SAC), hence actually satisfies (AC)” [1, Remark 6.4].

2 Definitions

Write the coordinates of a bivector $z \in \Lambda^2(\mathbb{R}^4)$ as

$$z = (z_{12}, z_{13}, z_{14}, z_{23}, z_{24}, z_{34}).$$

It is simple precisely when

$$z_{12}z_{34} - z_{13}z_{24} + z_{14}z_{23} = 0.$$

Every unit simple bivector determines an unoriented plane in $\mathbb{G}(4, 2)$, with the two choices z and $-z$ giving the same plane.

For $p > 1$, let

$$\mathcal{G}_p(z) = \left(\sum_{i < j} |z_{ij}|^p \right)^{1/p}, \quad \Psi_p(T_z) = \mathcal{G}_p(z)$$

for unit simple z . If $K(z)$ is the skew-symmetric matrix with $K(z)_{ij} = z_{ij}$ for $i < j$, and $\phi_p(s) = \text{sgn}(s)|s|^{p-1}$, equation (6.2) of [1] gives

$$B_p(T_z) = \frac{K(\phi_p(z))K(z)^\top}{\mathcal{G}_p(z)^{p-1}}. \quad (1)$$

For a probability measure μ on $\mathbb{G}(4, 2)$, put

$$A_p(\mu) = \int_{\mathbb{G}(4, 2)} B_p(T) d\mu(T).$$

Here the normal dimension is $n = 2$. Condition (AC2) says that if $\dim \ker A_p(\mu) = 2$, then μ must be a Dirac mass [1, Definition 3.1].

3 Proof intuition

The source proves that every averaged matrix has rank at least two, so a failure of (AC) must occur exactly at rank two with a nontrivial measure. We choose two simple bivectors obtained from one another by interchanging a large and a small Plücker coordinate. Their B_p matrices have complementary block forms. The equal average is symmetric and has only two eigenvalues, each repeated twice. As p varies from the nonsmooth endpoint 1 to the Euclidean exponent 2, the smaller eigenvalue changes sign. At an interior exponent it is therefore zero with multiplicity two, giving exactly the forbidden non-Dirac equality case.

4 Counterexample theorem

Theorem 1. *There exists $p_0 \in (1, 2)$ such that the Plücker-coordinate ℓ^{p_0} integrand Ψ_{p_0} on $\mathbb{G}(4, 2)$ does not satisfy the atomic condition. More precisely, there are two distinct planes $T_x, T_y \in \mathbb{G}(4, 2)$ such that, for*

$$\mu = \frac{1}{2}(\delta_{T_x} + \delta_{T_y}),$$

one has $\dim \ker A_{p_0}(\mu) = 2$.

Proof. Fix $a, b > 0$ such that

$$a + b = 1, \quad a \neq b,$$

and set $t = \sqrt{ab/2}$. In Plücker-coordinate order define

$$x = (a, -t, t, t, t, -b), \quad y = (b, -t, t, t, t, -a). \quad (2)$$

Both are simple, because

$$x_{12}x_{34} - x_{13}x_{24} + x_{14}x_{23} = -ab + t^2 + t^2 = 0,$$

and the same calculation applies to y . They are unit bivectors since

$$|x|^2 = |y|^2 = a^2 + b^2 + 4t^2 = a^2 + b^2 + 2ab = (a + b)^2 = 1.$$

The planes T_x and T_y are distinct: if they agreed, their unit simple representatives would satisfy $y = \pm x$, which is impossible when $a \neq b$ and $t > 0$.

Fix $p > 1$ temporarily and abbreviate

$$\alpha = a^{p-1}, \quad \beta = b^{p-1}, \quad \theta = t^{p-1}, \quad \rho_p = \mathcal{G}_p(x)^{p-1} = \mathcal{G}_p(y)^{p-1} > 0.$$

The equality defining ρ_p follows because x and y have the same multiset of absolute coordinates. Applying (1) gives

$$\rho_p B_p(T_x) = \begin{pmatrix} d_1 & 0 & -r & -r \\ 0 & d_1 & -r & r \\ -s & -s & d_2 & 0 \\ -s & s & 0 & d_2 \end{pmatrix}, \quad (3)$$

and

$$\rho_p B_p(T_y) = \begin{pmatrix} d_2 & 0 & -s & -s \\ 0 & d_2 & -s & s \\ -r & -r & d_1 & 0 \\ -r & r & 0 & d_1 \end{pmatrix}, \quad (4)$$

where

$$d_1 = \alpha a + 2\theta t, \quad d_2 = \beta b + 2\theta t, \quad r = \alpha t + \theta b, \quad s = \beta t + \theta a.$$

Consequently

$$A_p(\mu) = \frac{1}{\rho_p} \begin{pmatrix} d & 0 & -c & -c \\ 0 & d & -c & c \\ -c & -c & d & 0 \\ -c & c & 0 & d \end{pmatrix}, \quad (5)$$

with

$$d = \frac{\alpha a + \beta b}{2} + 2\theta t, \quad c = \frac{(\alpha + \beta)t + \theta}{2}; \quad (6)$$

we used $a + b = 1$ in the formula for c .

The characteristic polynomial of the unnormalized matrix in (5) is

$$((\lambda - d)^2 - 2c^2)^2.$$

Thus its eigenvalues are

$$d - \sqrt{2}c \quad \text{and} \quad d + \sqrt{2}c,$$

each with multiplicity two. Define $D(p) = d - \sqrt{2}c$ for $p > 1$ and extend it continuously to $p = 1$. Since $\alpha, \beta, \theta \rightarrow 1$ as $p \downarrow 1$,

$$D(1) = \frac{1}{2} + 2t - \sqrt{2} \left(t + \frac{1}{2} \right). \quad (7)$$

The right side is strictly increasing as a function of t , and it vanishes at $t = 1/(2\sqrt{2})$. But $a \neq b$ and $a + b = 1$ imply $ab < 1/4$, hence $t < 1/(2\sqrt{2})$. Therefore $D(1) < 0$.

At $p = 2$, one has $\alpha = a$, $\beta = b$, and $\theta = t$, so

$$d = \frac{a^2 + b^2}{2} + 2t^2 = \frac{1}{2}, \quad c = t,$$

and hence

$$D(2) = \frac{1}{2} - \sqrt{2}t = \frac{1}{2} - \sqrt{ab} > 0. \quad (8)$$

By the intermediate value theorem there exists $p_0 \in (1, 2)$ such that $D(p_0) = 0$. At this exponent $d = \sqrt{2}c > 0$, so the matrix in (5) has eigenvalues $0, 0, 2d, 2d$. The positive normalization ρ_{p_0} does not change the kernel. Thus

$$\dim \ker A_{p_0}(\mu) = 2,$$

although μ is supported on the two distinct planes T_x, T_y and is not Dirac. This violates (AC2), proving the theorem. $\square$

Concrete numerical location. Taking $a = 4/5$ and $b = 1/5$, bisection of the explicit continuous function D gives

$$p_0 \approx 1.1593840621.$$

This decimal is a reproducibility check only; the proof uses the strict signs in (7)–(8), not numerical computation.

5 Verification and limitations

The script `code/verify_counterexample.py` independently reconstructs the matrices from (1), checks the Plücker and norm identities, factors the characteristic polynomial, and bisects the concrete root. The computational output is not used as a proof.

The theorem gives a complete negative answer to the universal all- p claim suggested in Remark 6.4. It does not classify the exponents for which (AC) holds. It also does not contradict Theorem 6.1 of [1]: that theorem establishes (AC1) for every $p \in (1, \infty)$, whereas the construction violates only the equality-rigidity clause (AC2).

6 Bounded novelty check and review recommendation

On 19 July 2026, the run’s four lightweight indexes had no match for arXiv:2011.09922 or the core scalar-atomic/Plücker ℓ^p terms. Primary arXiv searches used the exact phrases “scalar atomic condition”, “uniform scalar atomic condition”, and “Plücker atomic condition”, together with the source title and authors. No prior resolution of Remark 6.4 was found. The closest later primary sources inspected were arXiv:2401.10858v2 (a necessary strict-polyconvexity condition for (AC)) and arXiv:2601.10647v2 (codimension-one ℓ^p anisotropies in a different Michael–Simon result); neither treats this two-plane $\mathbb{G}(4, 2)$ obstruction. This bounded search is not an exhaustive novelty guarantee.

Human review should first compare (1) with equation (6.2) of the source, then audit the signs in (3)–(4). The remaining kernel and intermediate-value argument is elementary.

References

- [1] A. De Rosa and R. Tione, *Regularity for graphs with bounded anisotropic mean curvature*, Invent. Math. 230 (2022), 463–507; arXiv:2011.09922v2.